

Object Oriented Programming

Pass Task 2.4: Case Study — Iteration 1: Identifiable Object

Overview

Object-oriented programming makes best sense with larger programs. The case study will be your opportunity to create a larger program and better understand how the object-oriented approach can make it easier to create complex software solutions.

Purpose: Practice interpreting UML class diagrams and writing unit tests.

Task: Understand the case study program and implement iteration 1.

The task contains personalized requirements.

Deadline: Due by the end of week three, **Fri, 16 August 2024 (Firmed)**.

Submission Details

All students have access to the Adobe Acrobat tools. Please print your solution to PDF and combine it with the screenshots taken for this task.

- Program source code
- Test source code
- Screenshot of unit tests passing

Instructions

1. Review the *Case Study Requirements* document and implementation plan included in the task resources. It outlines what you need to create.
2. For this week aim to complete Iteration 1.

Note: At this point there will not be a "program" as such, just a set of unit tests that help demonstrate that your solution is moving towards completion.

Once your tests are working correctly get a screenshot of the tests passing and submit them along with the code.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Swin_Adventure._2
{
    public class IdentifiableObject
    {
        private List<string> _Identifiers;

        public IdentifiableObject(string[] idents)
        {
            _Identifiers = new List<string>(idents);
        }
        public bool AreYou(string id)
        {
            return _Identifiers.Contains(id.ToLower());
        }
        public string FirstId
        {
            get
            {
                if (_Identifiers.Count == 0)
                {
                    return "";
                }
                else
                {
                    return _Identifiers[0];
                }
            }
        }
    }
}
```

```

    public void AddIdentifier(string id)
    {
        _Identifiers.Add(id.ToLower());
    }
    public void PrivilegeEscalation(string pin)
    {
        string studentid = "103995219";
        if (studentid.Substring(studentid.Length - 4) == pin)
        {
            _Identifiers[0] = "8";
        }
    }
}
}
using Swin_Adventure._2;

namespace IdentifiableObjectTest
{
    public class Tests
    {
        private IdentifiableObject _testObject;
        private string _testString;
        private string[] _testArray;

        private IdentifiableObject _testObject_emp;
        private string _testString_emp;
        private string[] _testArray_emp;

        [SetUp]
        public void Setup()
        {
            _testString = "ANH";
            _testArray = new string[] { "Anh", "Bob" };
            _testObject = new IdentifiableObject(_testArray);
            _testObject.AddIdentifier(_testString);

            _testString_emp = "";
            _testArray_emp = new string[] { };
            _testObject_emp = new IdentifiableObject(_testArray_emp);
            _testObject_emp.AddIdentifier(_testString_emp);
        }

        [Test]
        public void TestAreYou()
        {
            Assert.IsTrue(_testObject.AreYou(_testString));
        }
    }
}

```

```

    }

[Test]
public void TestNotAreYou()
{
    Assert.IsFalse(_testObject.AreYou("Max"));
}

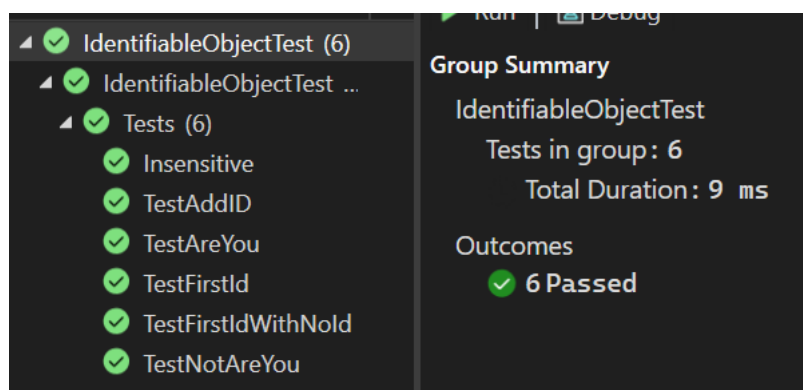
[Test]
public void Insensitive()
{
    Assert.IsTrue(_testObject.AreYou("ANH"));
}

[Test]
public void TestFirstId()
{
    Assert.AreEqual("Anh", _testObject.FirstId);
    Assert.AreNotEqual("max", _testObject.FirstId);
}

[Test]
public void TestFirstIdWithNold()
{
    Assert.AreEqual("", _testObject_emp.FirstId);
}

[Test]
public void TestAddID()
{
    _testObject.AddIdentifier("Max");
    _testObject.AddIdentifier("Andrew");
    Assert.IsTrue(_testObject.AreYou("Max"));
    Assert.IsTrue(_testObject.AreYou("Andrew"));
}
}
}

```



Assessment Criteria

Make sure that your task has the following in your submission:

- The “Universal Task Requirements” (see Canvas) have been met.

